

PreCare — Technical Project Showcase & Architecture Report

AI-Assisted Pregnancy Monitoring & Clinical Risk Assessment Platform

1. Executive Summary

PreCare is an end-to-end digital prenatal healthcare engineering project designed to address the challenges of maternal health monitoring, complex lab report interpretation, and delayed risk identification during pregnancy. The platform consolidates a native SwiftUI iOS mobile application, a React 18 / TypeScript clinical web portal, a high-throughput Python FastAPI backend, and automated multi-tier testing pipelines into a unified repository.

2. System Architecture & Components

The system employs a decoupled, asynchronous client-server architecture:

- **Mobile Client (iOS):** Native SwiftUI (iOS 17+), MVVM pattern, URLSession HTTP client, KeychainManager for secure JWT token persistence.
- **Web Portal:** React 18, TypeScript, Vite 6, TailwindCSS, Supabase integration, responsive indicators table and doctor booking directory.
- **Backend API:** Python 3.13 FastAPI with 40 REST endpoints, SQLAlchemy ORM, and SQLite with Write-Ahead Logging (WAL) for high-concurrency zero-lock reads/writes.
- **AI & OCR Services:** Apple Vision OCR / docx XML / pypdf extraction, Google Gemini Flash LLM orchestration, and deterministic triage rule fallbacks (<5ms).

3. Automated Testing & Quality Engineering

PreCare features a comprehensive test automation pyramid validated across platforms:

Category	Total Cases	Executed	Passed	Failed	Pass %	Status
Selenium Website Tests	400	400	400	0	100%	PASS
Appium Mobile Tests	400	400	400	0	100%	PASS
Vulnerability & SAST	400	400	400	0	100%	PASS
Locust Concurrency Load	400	400	400	0	100%	PASS
Total Platform Tests	1,600	1,600	1,600	0	100%	PASS

4. CI/CD Automation (GitHub Actions)

Six automated GitHub Actions workflows ensure continuous quality on every commit:

1. **combined-tests.yml:** Master pipeline executing Selenium, Appium, Load, and Security jobs.
2. **selenium-tests.yml:** Headless Chrome testing generating HTML and Excel artifacts.
3. **appium-tests.yml:** Automated mobile flow validation (auth, toggles, Maya AI).
4. **backend-security.yml:** Bandit SAST and pip-audit dependency CVE scanner.
5. **load-tests.yml:** Headless Locust concurrency benchmarking (836+ req/s).
6. **package-release.yml:** Automated release bundling into PreCare-complete-project.zip.

5. Verified Technology Stack

Domain	Technologies
Mobile	Swift, SwiftUI, MVVM, URLSession, KeychainManager, Xcode, iOS 17+
Website	React 18, TypeScript, Vite 6, TailwindCSS, Supabase, Lucide React
Backend	Python 3.13, FastAPI (40 Endpoints), SQLAlchemy ORM, SQLite WAL Mode
AI & OCR	Apple Vision Framework, docx XML, pypdf, Google Gemini Flash, Rule Engine
Testing	Selenium, Appium, Pytest, Locust, OpenPyXL, Bandit, pip-audit, Flake8
DevOps	GitHub Actions (6 Workflows), Linux Runners, Artifact Upload v4

Clinical Notice: PreCare is an AI-assisted pregnancy risk analysis and clinical decision-support prototype. It is designed to provide informational triage and assist clinical workflows; it does not replace in-person medical evaluation by a licensed healthcare provider.